Matias Barandiaran

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Education

1. Lancaster University – Leipzig, Germany (October 2022 – July 2025)
    a. Bachelor of Science in **Computer Science**.
    b. University ranked **#132** worldwide for Computer Science (QS by subject) in 2025.
    c. First Class Honours (**UK**: 91.2% ≈ **GPA**: 4.00/4.00).
    d. Ranked 1st in my graduating class.
2. International Baccaleaurate – Lima, Peru (March 2020 – February 2022)
    a. Bilingual Diploma.
    b. Cohort's top 5% (**GPA**: 3.9/4.0).

Awards / Honors:

1. **Chancellor's Medal** – Lancaster University (2025): The university's most prestigious award, presented annually to graduating students who demonstrate exceptional merit in their field.
2. **Highest Average Grade** – Lancaster University (2025): Cohort's highest average grade across all three years of the BSc Computer Science program.
3. **Outstanding Dissertation** – Lancaster University (2025): Obtained the highest achievable grade (100%) for the BSc Computer Science dissertation: *Growing Reservoirs with Developmental Graph Cellular Automata* (2025). **Strong Accept** for presentation at peer-reviewed International Conference on Artificial Life 2025 in Kyoto, Japan.
4. **EASST Best Paper Award** – Charles University (2025): Best ETAPS (International Joint Conferences On Theory and Practice of Software) 2025 paper related to the systematic and rigorous engineering of software and systems.

Work Experience

1. Technical University of Munich – Munich, Germany (July 2024 – August 2025)
    a. **Research Assistant** (Hiwi) | On-site
        i. Hosted by the Gagneur Lab (Computational Molecular Medicine) in affiliation with Helmholtz Munich.
        ii. Worked on rare variant association testing using deep learning and data-driven burden scores (DeepRVAT), utilizing **Python**, **R**, **PyTorch**, and **Snakemake**.
2. Martin Luther University of Halle-Wittenberg – Halle, Germany (July 2024 – September 2024)
    a. **Research Intern** | Remote
        i. Hosted by the Faculty of Medicine.
        ii. Worked on novel graph-level anomaly detection via latent space searching and generative adversarial networks (gladGAN) on molecular datasets, utilizing **Python** and **Pytorch**, and **scikit-learn**.
3. Lancaster University Leipzig – Leipzig, Germany (October 2023 – June 2024)
    a. **Teaching Assistant** | On-site
        i. Laboratory assistant and demonstrator for the Digital Systems module.
        ii. Delivered engaging and informative instructional sessions to facilitate student understanding on low-level programming with **MIPS** assembly.
4. Charles University – Prague, Czechia (August 2023 – September 2023)
    a. **Research Intern** | On-site
        i. Hosted by the Faculty of Mathematics and Physics.
        ii. Contributed to: *Unsatisfiability Proofs for Horn Solving* (2025). *ETAPS best paper award*
        iii. Designed production of *Alethe* verifiable proofs for the GOLEM Constrained Horn Clauses solver, utilizing **C++**.
5. Freelance – Lima, Peru (August 2020 – September 2021)
    a. **Math Tutor** | Hybrid
        i. Student advisor and tutor for IB Mathematics SL and HL.
        ii. Guided senior high school students to achieve outstanding exam performance and substantial growth in their mathematical abilities.

Publications

1. Rodrigo Otoni, Martin Blicha, Matias Barandiaran, Patrick Eugster, Jan Kofroˇn, and Natasha Sharygina. Unsatisfiability proofs for horn solving. In the International Conference on Tools and Algorithms for the Construction and Analysis of Systems, pages 67–87. Springer, 2025.
2. Matias Barandiaran and James Stovold. Growing reservoirs with developmental graph cellular automata. arXiv preprint arXiv:2508.08091, 2025.

Projects

1. **Industrial Waste Sorting Arm** – Lancaster University (2024)
    a. Integrated an industrial robotic arm with object detection and image classification software for the purpose of automated waste sorting, utilizing **Python**, **PyTorch**, **C++**, **C**, and **OpenCV**.
    b. Project funded by Lancaster University's Global Advancement Fund.
2. **Global Game Jam** – Lancaster University (2023, 2024)
    a. Participated in 48-hour game development events using **Java** and **libGDX**.
    b. Collaborated in a team using **agile development** and **continuous integration** practices.
    c. Represented the Computer Science Society.

Leadership Activities

1. **Computer Science Society** – Lancaster University (2023 – 2025)
    a. **Founder**: Founded the computer science society at my institution.
    b. Dedicated to fostering a culture of innovation, diversity and problem-solving.
    c. Secured university funding for the purchase of robotics supplies (e.g. 3D printer, Arduinos, etc.).
2. **Global Advancement Fund** – Lancaster University (2024)
    a. **Organizer**: Co-Initiated and championed the approval process of Lancaster University's Global Advancement Fund (~€20k) to facilitate international collaboration between campuses worldwide.
3. **Qiskit Fall Fest** – Lancaster University (2023)
    a. **Lead Organizer:** Organized and hosted a Quantum Computing (QC) event powered by IBM.
    b. Gave an introductory talk on QC and held workshops with Qiskit material. Assembled a series of seminars with special guests like Enrique Solano (Co-CEO of Kipu Quantum).

Certifications

1. **Advanced Applications of Neural Networks and Deep Learning** – University of Oxford (2023)
    a. Worked with Generative Adversarial Networks, Variational Autoencoders, Deep Reinforcement Learning, Diffusion Models and Graph Neural Networks.
    b. Overall grade: A+.
2. **Artificial Intelligence and Machine Learning: Theory and Practice** – University of Oxford (2023)
    a. Worked with Deep Neural Networks and Convolutional Neural Networks.
    b. Explored the mathematics and theory behind linear models.
    c. Overall grade: A+.
3. **Hardware of a Quantum Computer** – Delft University (2023)
    a. Studied the building blocks of quantum computers.
    b. Explored solid-state qubits (Transmon, silicon spin, diamond NV center) and topological qubits.
    c. Learned how quantum gates are implemented across qubit types.

Additional Skills and Interests

1. **Languages**: Spanish (native), English (C2) ***TOEFL**: 114/120*, German (A2).
2. **Tools**: Python, Java, R, PyTorch, Qiskit, Git, C, C++, SQL, Linux, Snakemake, Qiskit, Docker, LXD, QEMU, OS-Ken.
3. **Concepts**: Rare Variant Association Testing, Data Pipelines, Computer Vision, Generative Modelling, Supervised Learning, Graph Anomaly Detection, Reinforcement Learning, Reservoir Computing, Network Protocols, Distributed Systems.
4. **Hobbies**: Game Development, Scuba Diving, Badminton, Videography.